Characterization and Comparative Durability Study of Glass/Vinylester, Basalt/Vinylester, and Basalt/Epoxy FRP Bars

Brahim Benmokrane[1]; Fareed Elgabbas[2]; Ehab A. Ahmed, M.ASCE[3]; and Patrice Cousin[4]

Abstract: This paper presents an experimental study that investigated the physical, mechanical, and durability characteristics of three different types of fiber-reinforced polymer (FRP) bars made of basalt and glass fibers with vinylester and epoxy resins, namely basalt/vinylester (B/V) FRP bars, glass/vinylester (G/V) FRP bars, and basalt/epoxy (B/E) FRP bars. First, their physical and mechanical properties were assessed. Then, a comparative durability study was performed on the three types of FRP bars under alkaline exposure simulating a concrete environment. The alkaline exposure was achieved by immersing the bars in an alkaline solution for up to 5,000 h at 60°C. Thereafter, the properties were assessed and compared with the unconditioned (reference) values. The test parameters were (1) fiber type (basalt or glass); (2) resin type (vinylester or epoxy); and (3) conditioning time (1,000, 3,000, and 5,000 h). The test results reveal that the G/V composite had the best physical and mechanical properties and lowest degradation rate after conditioning in alkaline solution. The B/E composite ranked second, while the B/V composite evidenced the lowest physical and mechanical properties and exhibited significant degradation of its physical and mechanical properties after conditioning. **DOI: 10.1061/(ASCE)CC.1943-5614.0000564.** *© 2015 American Society of Civil Engineers.*

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Introduction

Recently, continuous effort in development and innovation in the fiber-reinforced polymer (FRP) technology are devoted towards using new types of fibers, such as basalt fibers in addition to the commonly used glass and carbon fibers. Basalt-fiber-reinforced polymer (BFRP) is expected to provide benefits that are comparable or superior to other types of FRP while being significantly cost-effective (Parnas et al. 2007; Wang et al. 2012; Wei et al. 2010; Lopresto et al. 2011). Few studies, however, have investigated BFRP characteristics and durability performance under real and simulated harsh environmental conditions. Since FRP is a combination of two or more different materials (fiber and resin) with a distinct interface between them, the constituent materials maintain their separate identities in the composite. Yet, their combination produces properties and characteristics that are different from those of the constituents and highly dependent on the cohesion between the fibers and resin. The resin system acts as a matrix bonding the fibers together and spreading the load applied to the composite between each of the individual fibers. The resin system also protects the fibers from abrasion and impact damage as well as severe environmental conditions—such as water, salts, and alkalis—which affect the durability of FRP products (SP System 1998). Furthermore, numerous studies have investigated FRP products manufactured with vinylester resin to determine the effect of environmental conditions (water, salts, alkalis) on their physical and mechanical properties (Mouritz et al. 2004; Wang 2005; Zou et al. 2008; Robert et al. 2009). The durability of BFRP bars, however, has yet to be investigated and understood.

GFRPs have been used for decades, following the investigation of their long-term behavior and durability characteristics as well as the development of material specifications [ACI 440.6M (ACI 2008); CAN/CSA S807 (CSA 2010b)]. Most of the GFRP products available are manufactured with E or ECR glass fibers and epoxy or vinylester resin. These GFRP bars have been used in many demonstration projects and field applications, such as bridges (Benmokrane et al. 2004, 2006, 2007; Ahmed et al. 2014), parking garages (Benmokrane et al. 2012), water-treatment plants (Mohamed and Benmokrane 2014), and concrete pavement (Benmokrane et al. 2008). Basalt fibers, however, are mainly produced in Ukraine and, more recently, in China. Most of the associated research work has focused on developing and characterizing the various BFRP products.

Sim et al. (2005) investigated the durability and performance at elevated temperatures of basalt, glass, and carbon fibers. They reported that immersing basalt and glass fibers in an alkaline solution led to volume and strength loss through a surface reaction, whereas carbon fibers did not show any significant strength reduction. The basalt fibers, however, retained about 90% of their strength at

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ambient temperature after exposure at 600°C for 2 h. Parnas et al. (2007) conducted an experimental study to determine if BFRP composites were feasible, practical, and a beneficial alternative for transportation applications. They concluded that the chemical composition of basalt fibers is close to glass fibers, except that the basalt contains a high ratio of iron oxide, conferring its brown color. They found no significant differences in stiffness or strength between basalt-fabric-reinforced polymer composites and glass composites reinforced with a fabric of similar weave. Aging results indicate that the interfacial region in basalt composites might be more vulnerable to environmental damage than in glass composites. The basalt-epoxy interface, however, might also be more durable than the glass-epoxy interface in tension-tension fatigue, because basalt composites have a longer fatigue life. Wang et al. (2008) systematically studied the chemical durability and mechanical properties of basalt fiber and its epoxy-resin composites in alkaline solution for 3 months. The experimental results showed that, after exposure, the modulus of the BFRP was unaffected, but its strength decreased by 40%. Li et al. (2012) studied the durability and fatigue performances of basalt-fiber/epoxy FRP bars in hydrothermal and alkaline environments. The bare basalt fiber immersed in these environments, without resin protection, exhibited a serious degradation of tensile properties due to severe fiber corrosion, as revealed by scanning electron microscopy. In contrast, the basalt-fiber reinforcing bars showed much improved durability when subjected to the same conditions.

An extensive research project is being conducted at the Department of Civil Engineering at the University of Sherbrooke to assess the short-term and long-term characteristics of newly developed BFRP bars as a preliminary step in introducing these new materials into pilot projects and FRP design codes and material specification. The project includes five different types of BFRP bars and one type of BFRP tendon. The preliminary results of this project (Vincent et al. 2013; Elgabbas et al. 2013) confirm the feasibility of producing new BFRP bars for structural concrete elements with physical and mechanical properties meeting ACI 440.6M (ACI 2008) and CAN/CSA S807 (CSA 2010b) requirements.

This paper presents an experimental investigation aimed at assessing the characteristics of three different types of fiber-reinforced polymer (FRP) bars made with basalt and glass fibers in vinylester and epoxy resins, namely basalt/vinylester (B/V) FRP bars, glass/vinylester (G/V) FRP bars, and basalt/epoxy (B/E) FRP bars. It also aimed at assessing their long-term durability through conditioning in alkaline solution simulating a moist concrete environment at high temperature. The tests findings on the characteristics and performance of basalt-fiber-reinforced polymer (BFRP) bars made with vinylester and epoxy resins were compared to commonly used glass-fiber-reinforced polymer (GFRP) made with vinylester resin. The findings of this comparison will contribute to integrating basalt FRP into FRP standards and guides, such as ACI 440.1 R (ACI 2006), ACI 440.6M (ACI 2008), CAN/CSA S807 (CSA 2010b), CAN/CSA S6 (CSA 2010a), and CAN/CSA S806 (CSA 2012). It should be mentioned that, so far, basalt-FRP bars have not been included in design standards and specifications.

Experimental Program Outline

This investigation was conducted on three different development FRP bars that were 6 mm diameter: basalt/vinylester (B/V) FRP bars, glass/vinylester (G/V) FRP bars, and basalt/epoxy (B/E) FRP bars. Standard glass fiber was used to manufacture the G/V bars. The experimental work was divided into three phases. Phase I included the determination of physical properties, which were compared to those obtained after conditioning. Phase II included mechanical characterization, including transverse shear strength, flexural strength, flexural modulus of elasticity (stiffness), and apparent horizontal shear strength (interlaminar shear strength). The test results also served as references for residual strength after conditioning. Phase III included a preliminary durability study and long-term performance assessment of FRP bars immersed in alkaline solutions simulating concrete pore solution at 60°C for different times up to 5,000 h. The changes in the physical and mechanical characteristics were assessed by comparing the characteristics of the conditioned bars to those of the reference bars from Phases I and II. The effects of conditioning on the glass transition temperature (T_g) and chemical composition of the materials were also assessed with differential scanning calorimetry (DSC) and Fourier transform infrared spectroscopy (FTIR), respectively. In addition, the microstructure of FRP bars—both conditioned and unconditioned specimens—was investigated with scanning electron microscopy (SEM) to assess any changes or degradation.

Material Properties and Test Specimens

The B/V, G/V, and B/E FRP bars were manufactured from continuous basalt or E-glass fibers impregnated in vinylester or epoxy resins using the pultrusion process. The basalt fibers (produced by Kamenny Vek, Dubna, Russia) had a diameter of 10–22 μm and had surface treatment consisting of a silane coupling agent. The moduli of elasticity of these basalt fibers ranged from 85 to 95 GPa (Kamenny Vek 2014). The E-glass fibers were produced by Owens Corning and their moduli of elasticity was about 78 GPa (Owens Corning 2014).

The B/V and G/V bars had smooth surfaces, whereas the B/E bars were deformed with helical wrapping and sand coating (Fig. 1). The smooth B/V and G/V bars were manufactured using closed-die pultrusion, while the B/E bars were fabricated with open-die pultrusion to allow for placement of helical fibers on the bar surface. The nominal cross-sectional area of the three bars is 28.3 mm^2, while the actual (immersion) cross-sectional area is 27.0 mm^2 for the B/V and G/V bars and 31.0 mm^2 for the B/E bars, as reported by the manufacturer. The mechanical properties reported in the current study were calculated using the nominal cross-sectional area.

For this research, transverse shear [ASTM D7617 (ASTM 2011)], flexural [ASTM D4476 (ASTM 2009)], and short-beam shear testing [ASTM D4475 (ASTM 2008)] were selected as the appropriate mechanical tests. The FRP specimens were cut into 200, 160, and 72 mm lengths for testing transverse shear strength, flexural properties, and short-beam shear, respectively.

Furthermore, a pilot investigation was conducted to evaluate the effects of chemicals on the basalt fibers used in manufacturing the BFRP bars tested herein. The tests followed the Owens Corning (2011) guide for glass fibers and the results are reported.

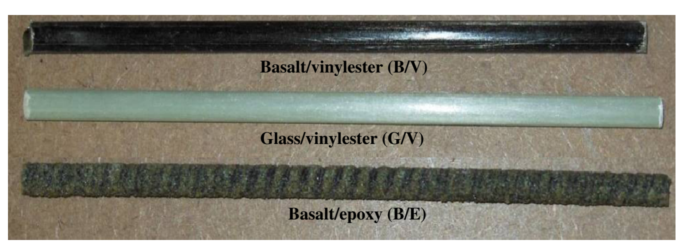


Fig. 1. Tested basalt and glass FRP bars

Table 1. Mass Loss (% by Weight) of Basalt Fibers after Conditioning in Water, Acidic, Saline, and Alkaline, Solutions at 96°C

Conditioning period	Mass loss (% by weight)			
	Water	Acidic	Saline	Alkaline
1 day	0.0	4.8	0.0	0.4
7 days	0.3	5.9	0.2	0.9

Chemical Resistance of Basalt Fibers

In parallel to the main investigation conducted herein on characterization and comparative durability assessment, chemical resistance tests were conducted on the basalt fibers (the same fibers used in manufacturing the tested BFRP) using the test procedure published by Owens Corning (2011) for evaluating glass fiber chemical resistance. Fibers were heated to 540°C overnight to remove any sizing and provide a proper clean surface for chemical-resistance investigation. Samples were then cut and carefully weighed before immersion in different corrosive aqueous solutions at 96°C for 1 and 7 days. The solutions were deionized water, an acidic solution (10% HCl), a saline solution (10% NaCl), and an alkaline solution (3.2 g NaOH per liter). After conditioning, the samples were thoroughly washed, dried, and weighed again. Table 1 presents the mass losses after 1 and 7 days of conditioning in deionized water, acidic solution, saline solution, and alkaline solution. The mass losses were less than 1% (wt) after immersion in water, alkaline solution, and saline solution, whereas the acidic solution yielded a mass loss ranging from 5 to 6% (wt). It should be mentioned that the color of the acidic solution turned green after conditioning, which may indicate some leaching of iron oxide.

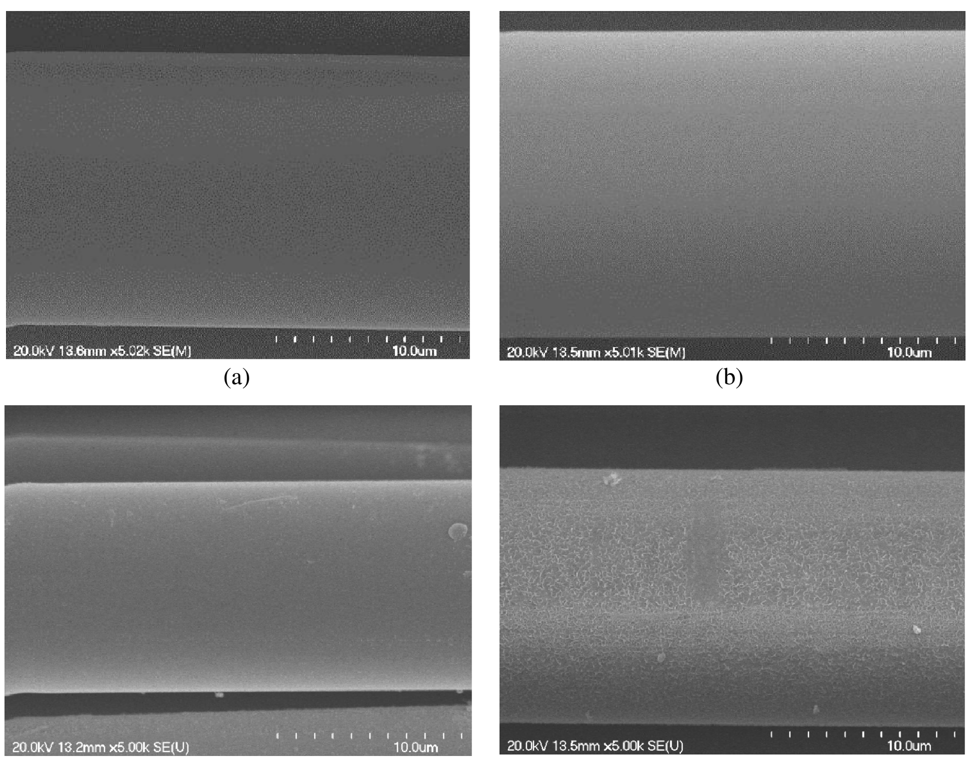
Furthermore, the conditioned fibers were then analyzed using SEM to detect any changes in microstructure. Fig. 2 presents typical micrographs obtained on unconditioned (reference) and conditioned samples (acidic, alkaline, and saline solutions). The basalt fiber conditioned in deionized water was similar to the reference [Figs. 2(a and b)]. The conditioning in the acidic and saline solutions did not visually affect the material [Fig. 2(c)]. On the other hand, the basalt fiber conditioned in the alkaline solution showed tiny signs of corrosion, i.e., the presence of chemical residue on the surface [Fig. 2(d)].

Testing, Results, and Discussion

Phase I: Physical Characterization

The physical properties for the reference (unconditioned) FRP bars were determined according to ACI (2008) and CSA (2010b). Table 2 lists the physical properties of the unconditioned FRP bars, where the glass transition temperature (T_g) was determined with differential scanning calorimetry (DSC) [ASTM D3418 (ASTM 2012b)], as shown in Fig. 3.

As shown in Table 2, the B/V and G/V FRP bars had approximately the same fiber content (83.4 and 82.3% by weight, respectively), while the B/E FRP had the lowest fiber-content ratio (78.9 wt%). Considering the density of basalt and glass fibers (2.67 and 2.6 g/cm^3, respectively) and assuming the resin density is 1.1 g/cm^3, the fiber volume fractions for B/V, G/V, and B/E are 67.4, 66.3, and 60.7%, respectively. The average cure ratios and transverse coefficients of thermal expansion of the tested bars were around 98.0 ± 1.0 and $21.8 \pm 0.8 \times 10^{-6}$°$C^{-1}$, respectively, without significant differences between the three tested bars.



(a) (b) (c) (d)

Fig. 2. SEM micrographs of basalt fibers: (a) before conditioning (reference); (b) after conditioning in acidic solution; (c) after conditioning in saline solution; (d) after conditioning in alkaline solution

Table 2. Physical Properties of the Reference FRP Bars

Property	FRP bar type B/V	G/V	B/E
Fiber content by weight (%)	83.4	82.3	78.9
Fiber content by volume (%)	67.4	66.3	60.7
Cure ratio (%)	98	97	99
Transverse CTE, ($\times 10^{-6}$°C^{-1})	21.1	22.7	21.7
Glass transition temperature, T_g (°C)	103	102	116
Moisture uptake (%)	0.41	0.21	0.47

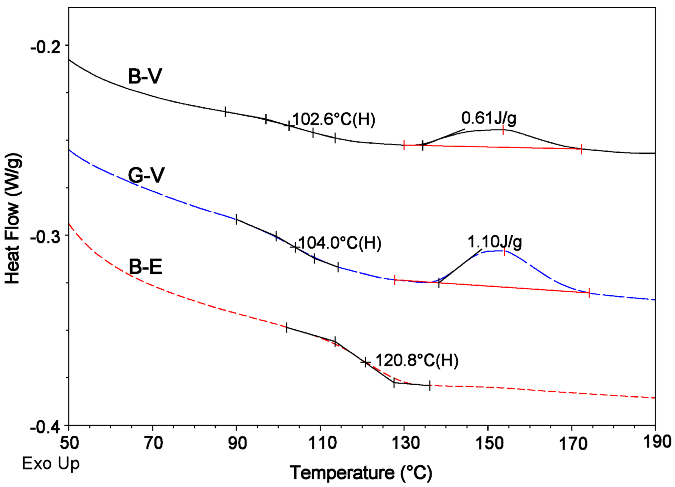


Fig. 3. DSC scans for glass transition temperature (T_g)

On the other hand, significant differences were observed for T_g and moisture uptake. The B/V and G/V FRP bars returned T_g values of 103 and 102°C, respectively, while the B/E FRP bars showed a T_g value of 116°C. On the other hand, the B/V and B/E bars had water-uptake ratios of 0.41 and 0.47%, respectively, while the G/V bars had a moisture-uptake ratio of 0.21% (approximately 50% of the moisture uptake of basalt FRP).

Phase II: Mechanical Characterization

The mechanical characterization included testing of representative FRP bars to determine their transverse shear strength, in accordance with ASTM D7617 (ASTM 2011); interlaminar shear (short-beam test), in accordance with ASTM D4475 (ASTM 2008); and flexural strength and flexural modulus of elasticity, in accordance with ASTM D4476 (ASTM 2009). Tensile-strength testing was not performed, since the specimens provided by the manufacturer were too short. In addition, the authors believe that those tests can provide a comparative performance assessment of the three FRP products tested herein. Figs. 4–6 show the mechanical characterization tests and Table 3 lists the results. The following sections provide brief descriptions and interpretation of the results.

Transverse-Shear Strength Test

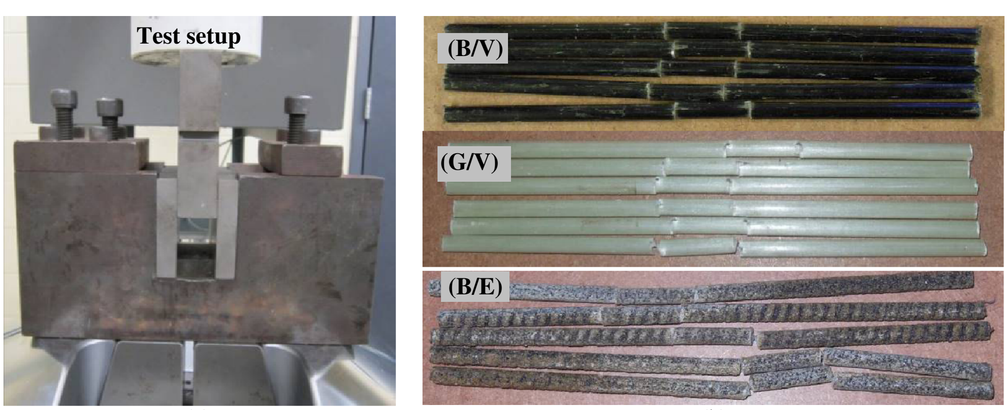
Transverse shear is the major structural force on dowels in jointed pavements or on stirrups in concrete beams. Transverse-shear tests were conducted according to ASTM D7617 (ASTM 2011) to characterize the FRP bars. The setup consisted of a $230 \times 100 \times 110$ mm steel base equipped with lower blades spaced at 50 mm face to face, allowing for the double transverse-shear failure of the specimen caused by an upper blade, as shown in Fig. 4.

For each type of FRP bar, six unconditioned specimens 200 mm in length were tested under laboratory conditions with an MTS 810 (MTS Systems Corporation, Eden Prairie, Minneapolis) testing machine equipped with a 500-kN load cell. A displacement-controlled rate of 1.3 mm/min was used during the test, which yielded between 30 and 60 MPa/min until specimen failure. The loading was done without subjecting the test specimens to any shock. The transverse-shear strength was calculated with Eq. (1)

$$\tau_u = \frac{P_s}{2A} \tag{1}$$

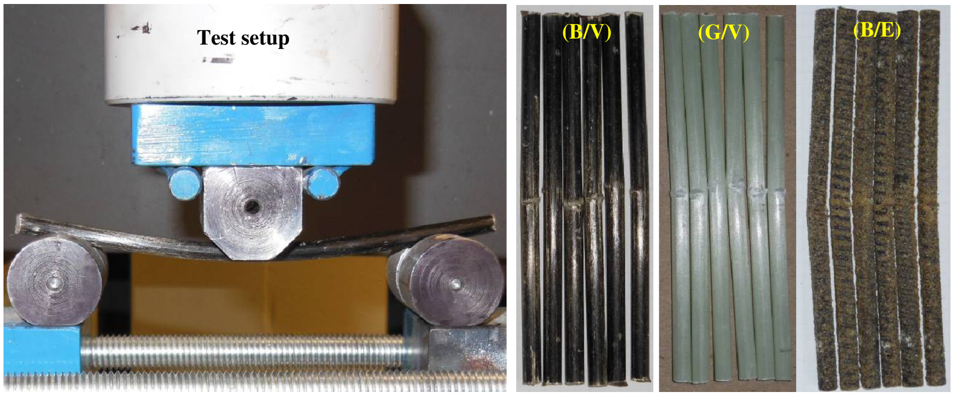
where τ_u = transverse-shear strength (MPa); P_s = failure load (N); and A = cross-sectional area of the FRP bar (mm^2).

Table 3 shows that the transverse-shear strengths of the B/V and G/V FRP bars were 263 ± 6 and 272 ± 1 MPa, respectively. The B/E FRP bars have the highest value of transverse-shear strength (283 ± 16 MPa). It is worth mentioning, however, that, although the transverse-shear strength results mainly from the resin, the fiber and the fiber/resin interface also play a role (Montaigu et al. 2013). The ratios between the shear strengths of the B/V and G/V and that of B/E bars were 0.93 and 0.96, respectively. The results indicate that the epoxy resin led to higher transverse-shear strength than the vinylester resin, but the standard deviation was high. Moreover, these values met the requirements of CSA (2010b), which specifies a minimum transverse-shear strength of 160 MPa for FRP bars.



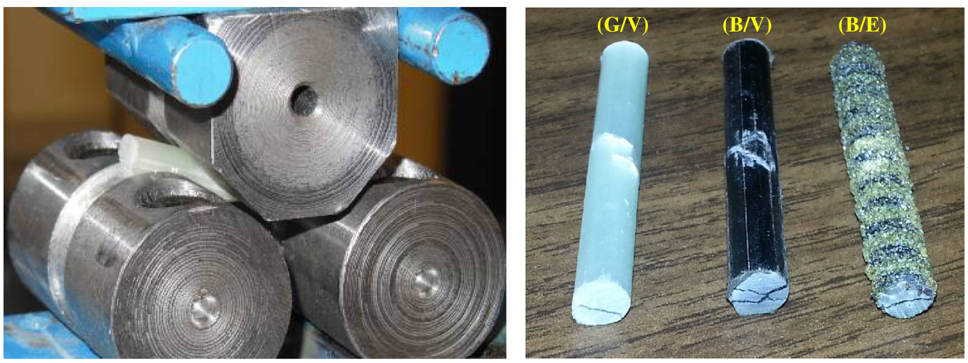
(a) (b)

Fig. 4. Setup for transverse-shear test and typical shear failure mode: (a) test setup; (b) failure mode



(a) (b)

Fig. 5. Setup for flexural testing and typical failure mode (compression failure): (a) test setup; (b) failure mode



(a) (b)

Fig. 6. Setup for short-beam testing and typical failure mode: (a) test setup; (b) failure mode

Three-Point Flexural Test

Flexural testing is especially useful for quality control and specification purposes. Flexural properties may vary with specimen diameter, temperature, weather conditions, and differences in rates of straining. The flexural properties obtained with this test method—ASTM D4476 (ASTM 2009)—cannot be used for design purposes but are appropriate for the comparative testing of composite materials. The test was conducted on specimens 160 mm long over a simply supported span equal to 20 times the bar diameter, as shown in Fig. 5. Six unconditioned specimens were tested under laboratory conditions as references for each type on an MTS 810 testing machine equipped with a 500-kN load cell. The specimens were loaded at the midspan with a circular nose; the specimen ends rested on two circular supports that allowed the specimens to bend. A displacement-controlled rate of 3.0 mm/min was used during the test. The rate of loading was done without subjecting the test specimen to any shock. The applied load and deflection were recorded during the test using a data acquisition system monitored by a computer.

Table 3. Mechanical Properties of the Reference FRP Bars

Bar type	τ_u (MPa)	f_u (MPa)	E (GPa)	ε_u (%)	S_u (MPa)
B/V	263 ± 6	$1{,}265 \pm 40$	60 ± 1	2.1	39 ± 3
G/V	272 ± 1	$1{,}555 \pm 123$	62 ± 1	2.5	50 ± 1
B/E	283 ± 16	$1{,}501 \pm 176$	41 ± 2	3.6	46 ± 3

Note: τ_u = transverse-shear strength (ASTM 2011); f_u = flexural strength in the outer fibers at midspan (ASTM 2009); E = flexural modulus of elasticity in bending (ASTM 2009); ε_u = maximum outer fiber strain (ASTM 2009); S_u = apparent horizontal shear (interlaminar) strength (ASTM 2008).

Flexural strength of tested FRP specimens was calculated with Eq. (2). Flexural modulus of elasticity (stiffness) is the ratio, within the elastic limit, of stress to corresponding strain. It was calculated with Eq. (3)

$$f_u = PLC/(4I) \quad (2)$$

$$E = PL^3/(48IY) \quad (3)$$

where f_u = flexural strength in the outer fibers at midspan (N/mm^2); P = failure load (N); L = clear span (mm); I = moment of inertia (mm^4); C = distance from the centroid to the extremities (mm); E = flexural modulus of elasticity in bending (N/mm^2); and Y = middle-span deflection at load P (mm).

The maximum outer fiber strain (ε_u) was calculated from Eq. (4)

$$\varepsilon_u = f_u/E \quad (4)$$

Table 3 provides the three-point flexural strength, flexural modulus of elasticity, and ultimate outer fiber strain of the tested

FRP bars. The elastic behavior of all the specimens was maintained until flexural failure, at which point the specimens failed due to compression in the top fibers, as shown in Fig. 5. The B/V bars showed the lowest flexural strength (1,265 ± 40 MPa), while the G/V FRP bars recorded the highest (1,555 ± 123 MPa). The B/E bars recorded a flexural strength of 1,501 ± 176 MPa. On the other hand, there were no significant differences between the flexural modulus of elasticity of both the B/V and G/V bars, which was 60 ± 1 and 62 ± 1 GPa, respectively. Finally, the B/E bars recorded the lowest flexural modulus of elasticity compared to vinylester-resin-based bars (65% of the B/V bars).

Short-Beam Shear Test

In FRP bars manufactured with a pultrusion process in which the fibers are arranged unidirectionally and bonded with the polymer matrix, the horizontal stresses would be more conducive to inducing interface degradation than transverse-shear stresses (Park et al. 2008). The short-beam shear test was conducted on six specimens of each type of FRP bar according to ASTM D4475 (ASTM 2008) in order to calculate the interlaminar shear strength, which is governed by the fiber/matrix interface. The tests were carried out with a 500-kN MTS 810 testing machine. The distance between the shear planes was set to 6 times the nominal diameter of the FRP bars. Fig. 6 shows the test setup and typical modes of failure of the tested specimens. A displacement-controlled rate of 1.3 mm/min was employed during the test. The applied load was recorded with a computer-monitored data acquisition system.

The interlaminar shear strength, S_u, of the FRP bars was calculated from Eq. (5)

$$S_u = 0.849P/d^2 \tag{5}$$

where S_u = interlaminar shear strength (MPa); P = shear failure load (N); and d = bar diameter (mm).

The short-beam shear test revealed that the G/V bars showed the highest interlaminar shear strength (50 ± 1 MPa), followed by the B/E bars (46 ± 3 MPa) and the B/V bars (39 ± 3 MPa). The results confirm that the interface between the basalt fibers and vinylester resin (B/V FRP bars) was not as strong as that of the G/V and B/E bars. Table 3 shows the apparent horizontal shear strength of the tested FRP bars. It is worth mentioning that the high values of the interlaminar shear strength reveal a strong interface between

Table 4. Retention and Mechanical Properties of Conditioned FRP Bars

Fiber/resin	Conditioned period	τ_u (MPa)	Retention (%)	f_u (MPa)	Retention (%)	E (GPa)	Retention (%)	S_u (MPa)	Retention (%)
B/V (h)	1,000	229	87	1,249	98	57	95	38	96
	3,000	198	75	926	73	54	89	34	87
	5,000	176	67	792	63	51	84	31	78
G/V (h)	1,000	253	93	1,492	96	58	93	47	93
	3,000	262	96	1,554	100	55	89	48	97
	5,000	245	90	1,447	93	50	81	47	95
B/E (h)	1,000	278	98	1,259	84	40	96	46	99
	3,000	279	98	1,078	72	39	96	41	88
	5,000	257	91	919	61	36	86	40	86

Note: E = flexural modulus of elasticity in bending; f_u = flexural strength in the outer fibers at midspan; S_u = apparent horizontal shear (interlaminar) strength; ε_u = maximum outer fiber strain; τ_u = transverse-shear strength.

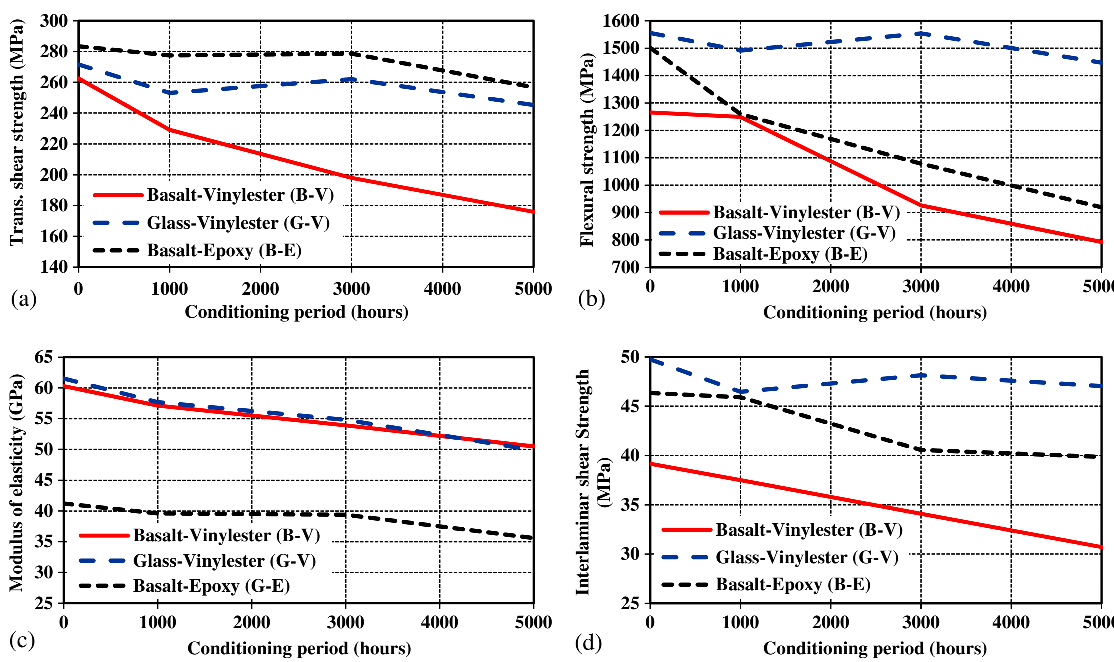


Fig. 7. Effect of conditioning in the alkaline solution at 60°C on the mechanical properties: (a) transverse-shear strength; (b) flexural strength; (c) flexural modulus of elasticity; (d) interlaminar shear strength

the resins and reinforcing fibers, which will be clarified in the SEM analysis to follow.

Phase III: Durability Study in Alkaline Solution

Conditioning of the FRP Bars in Alkaline Solution

Accelerated aging tests were conducted in accordance with ASTM D7705 (ASTM 2012a). The conditioning of FRP bars included the combined exposure to a harsh alkaline environment and elevated temperature. Immersion in an aqueous media (alkaline solution) at high temperature accelerates degradation. The alkaline solution was prepared to have a composition representative of the pore water inside portland cement concrete, specifically, 118.5 g of $Ca(OH)_2$, 0.9 g of NaOH, and 4.2 g of KOH per liter of deionized water. The solution had a pH of 12.6 to 13.0, which is representative of a mature concrete pore solution. The three types of FRP bars were immersed in this solution at 60°C for up to 5,000 h. The conditioning time started once the solution reached the prescribed temperature. Robert et al. (2009) reported that the increase in the degradation reaction rate was almost linear between room temperature and 50°C, whereas at higher temperatures (over 60°C), the increase in the degradation reaction rate was exponential. Therefore, to avoid any thermal degradation, the maximum conditioning temperature used in this study was 60°C, as specified in ASTM D7705 (ASTM 2012a).

The FRP specimens were placed in hermetically sealed stainless-steel containers to prevent excessive evaporation and the reaction of atmospheric CO_2 with calcium hydroxide. The containers were placed in an environmental chamber adjusted to the prescribed temperature (60°C) under isothermal conditions. The FRP bars were weighed and their diameters measured throughout the conditioning period to monitor water absorption and eventually characterize the mass and diameter changes. Observation revealed no changes in diameter during the conditioning periods. Six specimens of each type of FRP bars were removed from the solution and tested to determine their transverse-shear strength, interlaminar shear strength, flexural properties, and physical properties after 1,000, 3,000, and 5,000 h at 60°C. Similar to the mechanical characterization, the durability was assessed using tests for transverse-shear strength [ASTM D7617 (ASTM 2011)], interlaminate shear (short-beam test) [ASTM D4475 (ASTM 2008)], flexural strength, and flexural modulus of elasticity [ASTM D4476 (ASTM 2009)]. It is worth noting that degradation mechanisms in FRP bars are typically denoted as (1) fiber dominated; (2) matrix dominated; and (3) fiber-matrix interface dominated or combined mechanisms. Changes in mechanical properties determined by these tests are indicators of the three specific modes of degradation of the FRP constituent materials given earlier: fibers (flexural tests), resin (transverse and short-beam shear tests), and interface region (short-beam shear and flexural tests).

The results for the conditioned specimens were compared to those of the reference ones. The change in the different mechanical properties was selected as an indicator of a specific mode of environmental degradation of the FRP materials: fibers (flexural tests), resin (transverse-shear and short-beam shear tests), and the fiber/resin interface (short-beam shear and flexural tests).

Transverse-Shear Strength of the Conditioned FRP Bars

Table 4 shows the transverse-shear strength and strength-retention ratios of the tested FRP bars after 1,000, 3,000, and 5,000 h of immersion in the alkaline solution at 60°C. Table 4 indicates that the B/V bars were highly affected by accelerated aging with a transverse shear strength reduction of 33% after 5,000 h in the alkaline solution at 60°C, while the G/V and B/E bars were only slightly affected by accelerated aging with transverse-shear strength reductions of 10 and 9%, respectively. Fig. 7(a) shows the effect of the alkaline solution on the transverse shear strength after different exposure times. Contrary to the B/V bars, the G/V and B/E bars exhibited no significant reductions in the early stages (less than 3,000 h).

Flexural Strength of the Conditioned FRP Bars

Table 4 provides the flexural strength and strength-retention ratios of the tested FRP bars after 1,000, 3,000, and 5,000 h of immersion. Both the B/V and B/E bars had similar flexural strength reductions after 5,000 h (37 and 39%, respectively), while the G/V bars showed a slight reduction of 7%. These observations confirm that the bond between the basalt fibers and vinylester or epoxy resin—before and after conditioning—was lower than that between the glass fibers in the vinylester resin. Consequently, debonding occurring at the fiber/matrix interface made the fibers separate from the resin. Fig. 7(b) shows the effect of the alkaline solution on flexural strength. The lowest reduction rate was observed with the G/V bars, which yielded the lowest degradation at the interface. The high

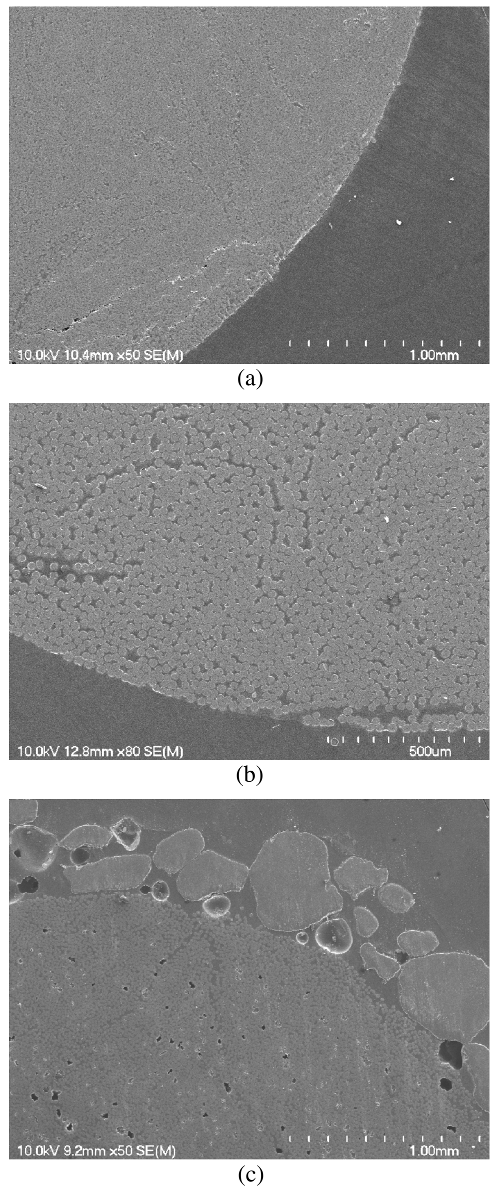


Fig. 8. Micrographs of the cross-section of the reference FRP bars: (a) B/V; (b) G/V; (c) B/E

degradation of the B/E bars after 1,000 h of conditioning resulted from the penetration of the alkaline solution through the initial voids. The B/V bars, however, returned an almost steady degradation rate between 1,000 and 5,000 h.

Flexural Modulus of Elasticity of the Conditioned FRP Bars

Table 4 gives the flexural modulus of elasticity and the retention ratio of the tested FRP bars after 1,000, 3,000, and 5,000 h of immersion. The three FRP bars showed no significant differences in flexural modulus of elasticity after 5,000 h. The reduction ranged from 14 to 19% as compared to the references. Fig. 7(c) illustrates the effect of the alkaline solution on the flexural modulus of elasticity where specimens exhibited steady reduction rate for all types of FRP bars.

Interlaminar Shear Strength of the Conditioned FRP Bars

Table 4 also shows the apparent horizontal shear (interlaminar shear) strength and strength retention ratios of the tested FRP bars after 1,000, 3,000, and 5,000 h of immersion. As for flexural testing, the G/V bars offered excellent stability and durability after immersion in the alkaline solution; they were followed by the B/E and B/V bars. The reduction ratios for the B/V, G/V, and B/E bars after 5,000 h were 22, 5, and 14%, respectively. Again, this observation confirmed the strong fiber/resin interface of the G/V bars, followed by the B/E and B/V bars. As evidenced from these results, the fiber/resin interface stands out as one of the most important issues in manufacturing basalt FRP. Fig. 7(d) shows the effect of the alkaline solution on the interlaminar shear strength, whereas the V-G FRP bars exhibited the lowest rate of degradation.

Microstructural Analysis of the Reference and Conditioned FRP Bars

SEM observations were performed to investigate microstructural changes in the FRP bars before and after conditioning. The specimens were cut, polished, and coated with a thin layer of gold/palladium in a vapor-deposit process. The analysis was carried out on a JEOL JSM-840 A microscope (JEOL, Akishima, Tokyo, Japan). Figs. 8–11 show the SEM micrographs of the reference and 5,000-h conditioned specimens.

SEM analysis of reference specimens (Figs. 8–11) indicates that the FRP bars made with vinylester resin (B/V and G/V) were nonporous but presented some debonding at the interface between the fibers and resin. This debonding was higher for the B/V bars than the G/V bars. Consequently, the B/V bars evidenced higher moisture uptake measured at saturation and higher degradation rate of mechanical properties compared to the G/V bars. On the other hand, the bars containing the epoxy resin (B/E) revealed a certain level of porosity, but offer better bonding between the fibers and resin.

SEM micrographs of 5,000-h conditioned specimens, also shown in Figs. 8–11, show no significant damage to the fiber surface, interface fiber/matrix, or microstructure of the G/V and B/E bars after aging. The specimens with initial debonding between the fibers and resin (B/V) seem to be affected by the conditioning, leading to more debonding at the fiber-resin interface. This justifies the lowest retention in the interlaminar shear strength, which was 67%.

Furthermore, SEM was also performed on the fracture zones of 1,000-h conditioned specimens after short-beam testing (Fig. 12) to investigate the mechanisms of failure at the interface fiber/matrix. The fiber surface of the G/V bars had more resin coverage [Fig. 12(b)] than the B/E [Fig. 12(c)] and B/V bars [Fig. 12(a)]. This observation corroborates the reduction ratio of the interlaminar shear strength and flexural strength after conditioning in the alkaline solution and characterizes the higher bonding of the glass fiber to the vinylester resin.

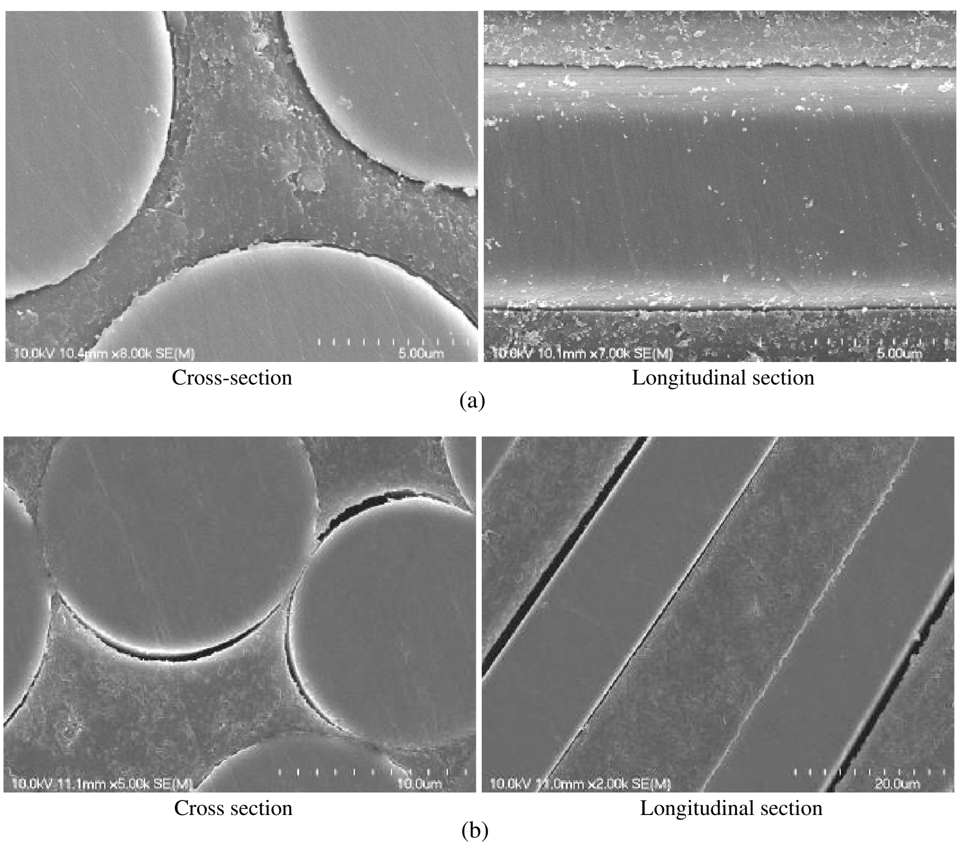
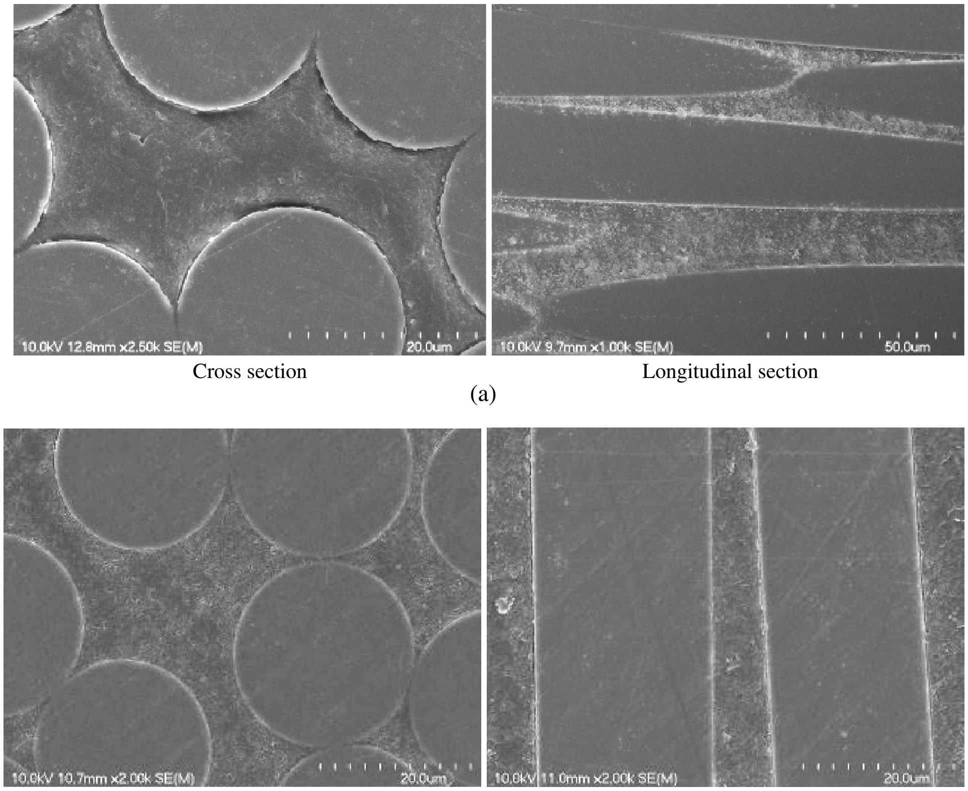


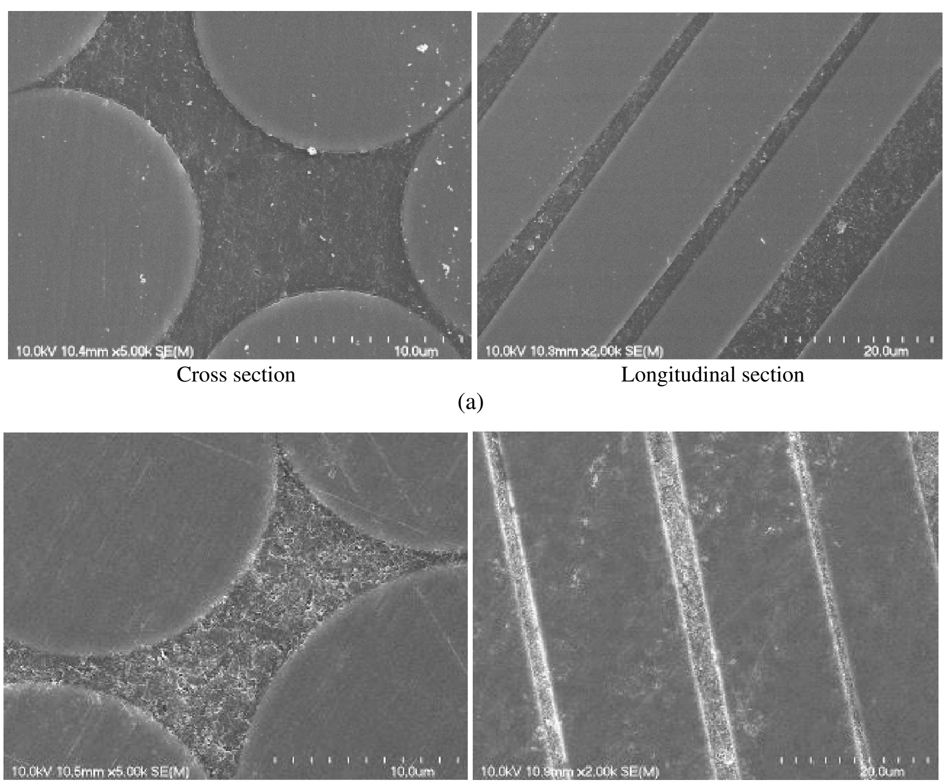
Fig. 9. Micrographs of the fiber/matrix interface of a B/V bar before and after conditioning: (a) before conditioning; (b) after conditioning



Cross section
Longitudinal section
(a)

Cross section
Longitudinal section
(b)

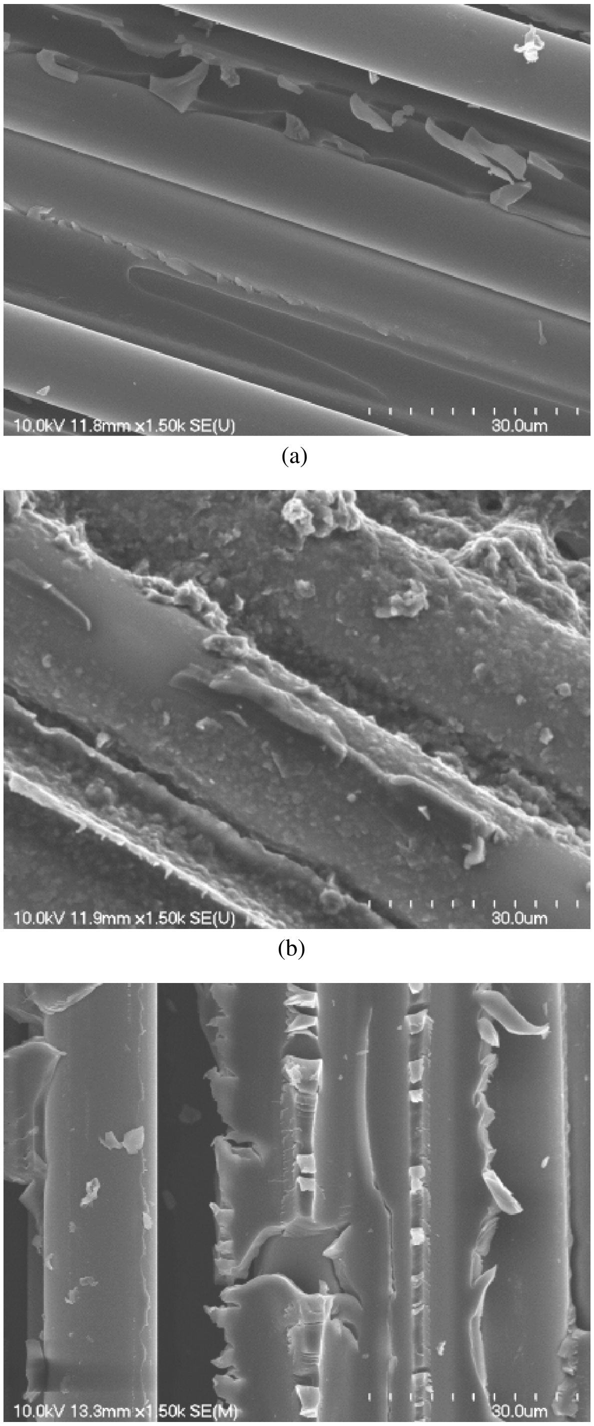
Fig. 10. Micrographs of the fiber/matrix interface of a G/V bar before and after conditioning: (a) before conditioning; (b) after conditioning



Cross section
Longitudinal section
(a)

Cross section
Longitudinal section
(b)

Fig. 11. Micrographs of the fiber/matrix interface of a B/E bar before and after conditioning: (a) before conditioning; (b) after conditioning



(c)

Fig. 12. Micrograph of bars conditioned in the alkaline solution for 1,000 h at 60°C (after interlaminar shear failure): (a) B/V; (b) G/V; (c) B/E

Glass Transition Temperature and Cure Ratio of the Conditioned FRP Bars

Differential scanning calorimetry (DSC) is used to obtain information about the thermal behavior and characteristics of polymeric materials and composites, such as the glass transition temperature (T_g) and curing process. In this study, 30–50 mg specimens from both unconditioned and conditioned specimens were sealed in aluminum pans and heated in a TA Instruments (New Castle, Delaware) DSC Q10 calorimeter to 200°C at a rate of 20°C/ min. The glass transition temperature (T_g) was determined in accordance with ASTM D3418 (ASTM 2012b).

Table 5. T_g, Cure Ratio, and Moisture Uptakes of Reference and Conditioned FRP Bars

	B/V		G/V		B/E	
Property	Reference	5,000 h	Reference	5,000 h	Reference	5,000 h
Cure ratio (%)	98	100	97	100	99	100
T_g (°C)	103	110	102	111	116	112
Moisture uptake (%)	0.41	0.57	0.21	0.26	0.47	0.79

Table 5 presents the T_g values for the reference and specimens conditioned for 5,000 h. The T_g of the conditioned B/V and G/V bars were slightly increased compared to the reference specimens, as a result of postcuring at high temperature. The B/E bars were almost fully cured (99%), where the T_g was slightly decreased with respect to the reference specimens. The B/E bars were made of epoxy resin, which is known to present a decrease in T_g when water is absorbed (plasticizing effect). The water absorption of the B/E bars was 0.79%.

Chemical Changes in Conditioned FRP Bars

FTIR was used to identify any chemical change/degradation after conditioning. FTIR spectra were recorded using an ABB-Bomen (ABB Bomem, Quebec, Canada) (MB series) spectrometer equipped with an attenuated total reflectance device. Thirty-two scans were routinely acquired with an optical retardation of 0.25 cm to yield a resolution of 4 cm^{-1}. The reference and 5,000-h conditioned specimens were investigated.

Fig. 13 shows the FTIR spectra of the unconditioned and conditioned FRP in the alkaline solution for 5,000 h at 60°C. Hydrolysis reaction results in the formation of new hydroxyl groups due to the alkaline attack of sensitive units, such as ester groups, with corresponding increases in the infrared band located between 3,200 and 3,600 cm^{-1}, which corresponds to the stretching mode of the hydroxyl groups in the vinylester and epoxy resins. In all the tested specimens, the hydroxyl peak did not show any significant changes, which means that the amount of hydroxyl groups present in the resins had not increased. This observation confirms that the epoxy and vinylester resins used did not degrade chemically during the immersion of the FRP bars in the alkaline solution at 60°C for 5,000 h.

Moisture Uptake at Saturation of the Conditioned FRP Bars

The moisture uptake at saturation of the reference and conditioned FRP bars was determined according to ASTM D570 (ASTM 2010). Three 75-mm-long specimens for each type of FRP bar were cut, dried, and weighed. They were then immersed in water at 50°C for 3 weeks. The samples were periodically removed from the water, dried, and weighed. The water content as a percentage of weight was calculated with Eq. (6)

$$W = 100(P_s - P_d)/P_d \tag{6}$$

where P_s and P_d are the weights of the bar in the saturated and dried states, respectively. The percentage of moisture uptake was calculated. The gain in mass was corrected to account for specimen mass loss due to possible dissolution phenomena. This correction was achieved by completely drying the immersed specimens in an

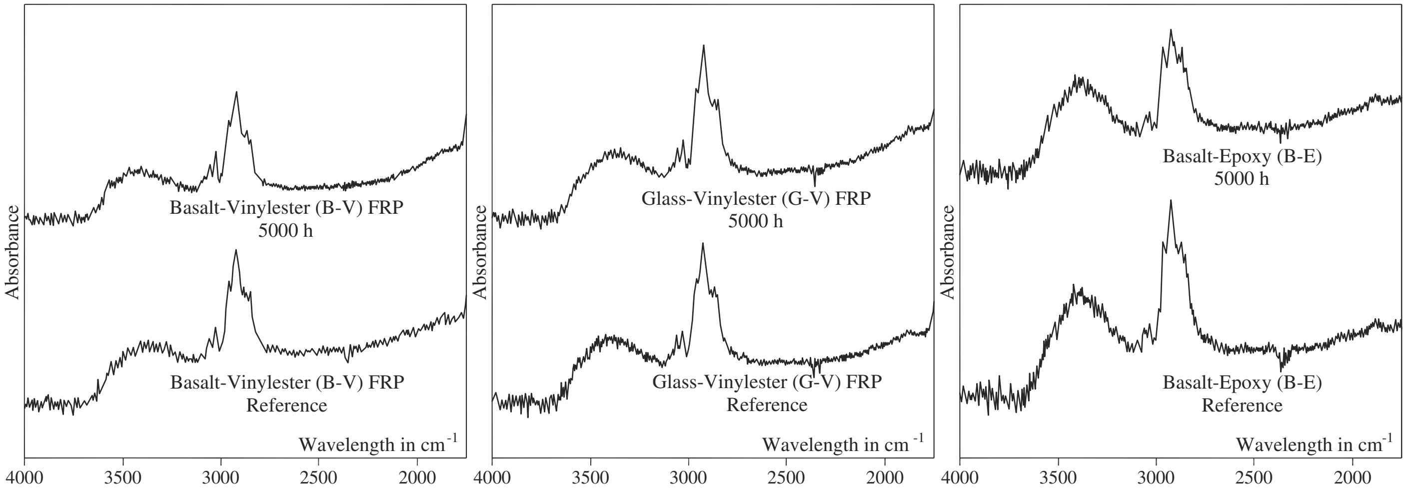


Fig. 13. FTIR spectra of reference and 5,000-h conditioned specimens

oven at 100°C for 24 h and comparing their masses to their initial masses. It should be mentioned that the mass loss may have occurred due to several causes: dissolution of soluble chemicals present on the surface, sand debonding in the case of sand-coated bars, and chemical degradation of one of the components, such as hydrolysis of the resin. In the presented case, the mass loss was less than 0.1%, which is too small to be related to any chemical degradation.

Moisture-uptake ratios at saturation for the reference specimens were 0.41, 0.21, and 0.47% for the B/V, G/V, and B/E FRP, respectively, while these ratios for the conditioned specimens were 0.26 ,0.57, and 0.79% for the B/V, G/V, and B/E FRP, respectively. The lowest water uptakes can be observed with G/V FRP, which is consistent with the lowest degradation of the fiber/resin interface as determined by DSC analysis and SEM observations (Figs. 8–11).

Summary and Conclusions

This investigation was conducted on three different development fiber-reinforced polymer (FRP) bars: basalt/vinylester (B/V) FRP bars, glass/vinylester (G/V) FRP bars, and basalt/epoxy (B/E) FRP bars. Based on the results of this research study, the following conclusions concerning the basalt and glass FRP bars made with vinylester and epoxy tested herein can be drawn:

1. The glass/vinylester FRP bars exhibited the best bond between the fibers and resin, flexural strength, flexural modulus of elasticity, and interlaminar shear strength, which is governed by the fiber/matrix interface. In addition, they showed the lowest moisture uptake.
2. Both the unconditioned basalt/vinylester and basalt/epoxy bars exhibited relatively high values of water uptake (twice that of the G/V bars), which is possibly related to the delamination debonding of the fiber/resin interface in the former and presence of some porosity in the second. The B/E bars were manufactured with a different manufacturing process—open-die pultrusion compared to closed-die pultrusion for the B/V bars—which seems to increase porosity.
3. The unconditioned B/V bars exhibited the lowest transverse-shear strength, flexural strength, interlaminar shear strength, and worst fiber/resin interface. The transverse-shear strength of the B/V bars was significantly affected by accelerated aging (reduced by 33% after 5,000 h), while the G/V and B/E bars were slightly affected by accelerated aging (reduced by 10 and 9%, respectively, after 5,000 h).
4. The flexural strength of the B/V and B/E bars was significantly affected by accelerated aging (reduced by 37% and 39%, respectively, after 5,000 h), while the G/V bars were slightly affected by accelerated aging (reduced by 7% after 5,000 h).
5. The interlaminar shear strength of the B/V and B/E bars was affected by accelerated aging (reduced by 22 and 14%, respectively, after 5,000 h), while the G/V bars were slightly affected by accelerated aging (reduced by 5% after 5,000 h). The fiber-resin interface plays a significant role in controlling the degradation due to conditioning.
6. Basalt and glass fibers were unaffected by conditioning since no degradation was observed in either the glass or basalt fibers. FTIR analysis confirmed that basalt and glass bars made with vinylester and epoxy resin had not been chemically degraded.
7. The vinylester FRP bars showed an increase in the T_g by about 8°C after conditioning due to postcuring (cure ratio of the reference specimens was 97.5%). On the other hand, the epoxy FRP bars showed a slight decrease in T_g after conditioning.
8. The basalt FRP bars showed an increase in moisture uptake after conditioning by 40% and 68% for the B/V and B/E bars, respectively. The epoxy resin absorbed more water than the vinylester. It should be mentioned, however, that their reference specimens showed very similar moisture uptake (0.41 and 0.47% for the B/V and B/E bars, respectively).

In summary, the glass/vinylester FRP bars showed superior durability in the alkaline environment at elevated temperature compared to the basalt/vinylester and basalt/epoxy bars. The results confirmed that the two types of fiber (glass and basalt) and resins (vinylester and epoxy) used in this study were not affected by the conditioning. The strength degradation observed for both types of basalt FRP bars was attributed to the sizing, which evidenced a poor bond between the resin and basalt fibers, as confirmed by SEM. The basalt FRP industry needs to improve this aspect to obtain basalt-FRP bars that perform as well as glass-FRP bars in a concrete environment. Further investigations into this relatively new material in civil engineering are needed.

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